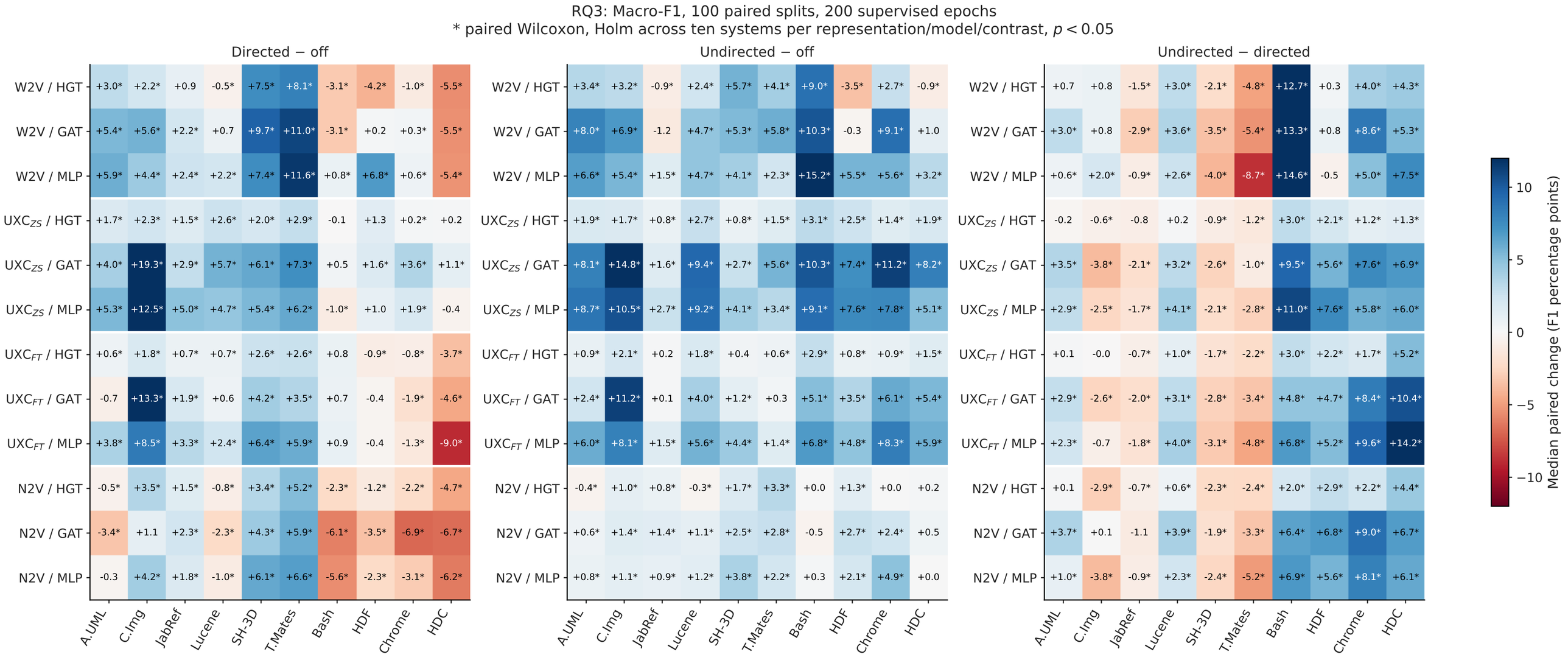
RQ3: Macro-F1, 100 paired splits, 200 supervised epochs
* paired Wilcoxon, Holm across ten systems per representation/model/contrast, $p < 0.05$
Directed − off
Undirected − off
Undirected − directed
W2V / HGT +3.0* +2.2* +0.9 -0.5* +7.5* +8.1* -3.1* -4.2* -1.0* -5.5*
W2V / GAT +5.4* +5.6* +2.2* +0.7 +9.7* +11.0* -3.1* +0.2 +0.3* -5.5*
W2V / MLP +5.9* +4.4* +2.4* +2.2* +7.4* +11.6* +0.8* +6.8* +0.6* -5.4*
UXC$_{ZS}$ / HGT +1.7* +2.3* +1.5* +2.6* +2.0* +2.9* -0.1 +1.3 +0.2* +0.2
UXC$_{ZS}$ / GAT +4.0* +19.3* +2.9* +5.7* +6.1* +7.3* +0.5 +1.6* +3.6* +1.1*
UXC$_{ZS}$ / MLP +5.3* +12.5* +5.0* +4.7* +5.4* +6.2* -1.0* +1.0 +1.9* -0.4
UXC$_{FT}$ / HGT +0.6* +1.8* +0.7* +0.7* +2.6* +2.6* +0.8 -0.9* -0.8* -3.7*
UXC$_{FT}$ / GAT -0.7 +13.3* +1.9* +0.6 +4.2* +3.5* +0.7 -0.4 -1.9* -4.6*
UXC$_{FT}$ / MLP +3.8* +8.5* +3.3* +2.4* +6.4* +5.9* +0.9 -0.4 -1.3* -9.0*
N2V / HGT -0.5* +3.5* +1.5* -0.8* +3.4* +5.2* -2.3* -1.2* -2.2* -4.7*
N2V / GAT -3.4* +1.1 +2.3* -2.3* +4.3* +5.9* -6.1* -3.5* -6.9* -6.7*
N2V / MLP -0.3 +4.2* +1.8* -1.0* +6.1* +6.6* -5.6* -2.3* -3.1* -6.2*
W2V / HGT +3.4* +3.2* -0.9* +2.4* +5.7* +4.1* +9.0* -3.5* +2.7* -0.9*
W2V / GAT +8.0* +6.9* -1.2 +4.7* +5.3* +5.8* +10.3* -0.3 +9.1* +1.0
W2V / MLP +6.6* +5.4* +1.5* +4.7* +4.1* +2.3* +15.2* +5.5* +5.6* +3.2*
UXC$_{ZS}$ / HGT +1.9* +1.7* +0.8* +2.7* +0.8* +1.5* +3.1* +2.5* +1.4* +1.9*
UXC$_{ZS}$ / GAT +8.1* +14.8* +1.6* +9.4* +2.7* +5.6* +10.3* +7.4* +11.2* +8.2*
UXC$_{ZS}$ / MLP +8.7* +10.5* +2.7* +9.2* +4.1* +3.4* +9.1* +7.6* +7.8* +5.1*
UXC$_{FT}$ / HGT +0.9* +2.1* +0.2 +1.8* +0.4 +0.6* +2.9* +0.8* +0.9* +1.5*
UXC$_{FT}$ / GAT +2.4* +11.2* +0.1 +4.0* +1.2* +0.3 +5.1* +3.5* +6.1* +5.4*
UXC$_{FT}$ / MLP +6.0* +8.1* +1.5* +5.6* +4.4* +1.4* +6.8* +4.8* +8.3* +5.9*
N2V / HGT -0.4* +1.0* +0.8* -0.3* +1.7* +3.3* +0.0 +1.3* +0.0 +0.2
N2V / GAT +0.6* +1.4* +1.4* +1.1* +2.5* +2.8* -0.5 +2.7* +2.4* +0.5
N2V / MLP +0.8* +1.1* +0.9* +1.2* +3.8* +2.2* +0.3 +2.1* +4.9* +0.0
W2V / HGT +0.7 +0.8 -1.5* +3.0* -2.1* -4.8* +12.7* +0.3 +4.0* +4.3*
W2V / GAT +3.0* +0.8 -2.9* +3.6* -3.5* -5.4* +13.3* +0.8 +8.6* +5.3*
W2V / MLP +0.6* +2.0* -0.9* +2.6* -4.0* -8.7* +14.6* -0.5 +5.0* +7.5*
UXC$_{ZS}$ / HGT -0.2 -0.6* -0.8 +0.2 -0.9* -1.2* +3.0* +2.1* +1.2* +1.3*
UXC$_{ZS}$ / GAT +3.5* -3.8* -2.1* +3.2* -2.6* -1.0* +9.5* +5.6* +7.6* +6.9*
UXC$_{ZS}$ / MLP +2.9* -2.5* -1.7* +4.1* -2.1* -2.8* +11.0* +7.6* +5.8* +6.0*
UXC$_{FT}$ / HGT +0.1 -0.0 -0.7* +1.0* -1.7* -2.2* +3.0* +2.2* +1.7* +5.2*
UXC$_{FT}$ / GAT +2.9* -2.6* -2.0* +3.1* -2.8* -3.4* +4.8* +4.7* +8.4* +10.4*
UXC$_{FT}$ / MLP +2.3* -0.7 -1.8* +4.0* -3.1* -4.8* +6.8* +5.2* +9.6* +14.2*
N2V / HGT +0.1 -2.9* -0.7* +0.6* -2.3* -2.4* +2.0* +2.9* +2.2* +4.4*
N2V / GAT +3.7* +0.1 -1.1 +3.9* -1.9* -3.3* +6.4* +6.8* +9.0* +6.7*
N2V / MLP +1.0* -3.8* -0.9* +2.3* -2.4* -5.2* +6.9* +5.6* +8.1* +6.1*
A.UML
C.Img
JabRef
Lucene
SH-3D
T.Mates
Bash
HDF
Chrome
HDC
Median paired change (F1 percentage points)
10
5
0
−5
−10